[This question paper contains 2 printed pages.]

Your Roll No...............

Sr. No. of Question Paper	:	**4686**	**E**
Unique Paper Code	:	32351402	
Name of the Paper	:	Riemann Integration and Series of Functions	
Name of the Course	:	**B.Sc. (H) Mathematics**	
Semester	:	IV	

Duration : 3 Hours   Maximum Marks : 75

Instructions for Candidates

1. Write your Roll No. on the top immediately on receipt of this question paper.
2. Attempt **two** parts from each question.

1(a) Let f be integrable on [a, b], and suppose g is a function on [a, b] such that $g(x) = f(x)$ except for finitely many x in [a, b]. Show g is integrable and $\int_a^b g = \int_a^b f$ (6)

(b) Show that if f is integrable on [a, b] then f^2 also is integrable on [a, b]. (6)

(c) (i) Let f be a continuous function on [a,b] such that $f(x) \geq 0$ for all $x \in [a, b]$. Show that if $\int_a^b f(x)\,dx = 0$ then $f(x) = 0$ for all $x \in [a, b]$ (3)

(ii) Give an example of function such that $|f|$ is integrable on [0,1] but f is not integrable on [0,1]. Justify it. (3)

2(a) State and prove Fundamental Theorem of Calculus I. (6.5)

(b) State Intermediate Value Theorem for Integrals. Evaluate $\lim_{x\to 0} \frac{1}{x}\int_0^x e^{t^2}\,dt$. (6.5)

(c) Let function $f: [0,1] \to R$ be defined as

$$f(x) = \begin{cases} x^2 & if\ x\ is\ rational \\ 0 & if\ x\ is\ irrational \end{cases}$$

Calculate the upper and lower Darboux integrals for f on the interval [0,1]. Is f *integrable on [0.1]* ? (6.5)

3(a) *Examine the convergence of the* improper integral $\int_0^\infty e^{-x}x^{n-1}dx$. (6)

(b) *Show that the improper integral* $\int_\pi^\infty \frac{Sin x}{x}dx$ is convergent but not absolutely convergent. (6)

P.T.O.

(c) Determine the convergence or divergence of the improper integral (6)

(i) $\int_0^1 \frac{dx}{x(\ln x)^2}$

(ii) $\int_1^\infty \frac{x\,dx}{\sqrt{x^3+x}}$

4(a) Show that the sequence

$$f_n(x) = \frac{nx}{1+nx}, \quad x \in [0,1], \quad n \in N$$

converges non-uniformly to an integrable function f on [0,1] such that

$$\lim_{n\to\infty} \int_0^1 f_n(x)dx = \int_0^1 f(x)dx \qquad (6.5)$$

(b) Show that the sequence $\{x^2 e^{-nx}\}$ converges uniformly on $[0,\infty)$. (6.5)

(c) Let $\langle f_n \rangle$ be a sequence of continuous function on $A \subset R$ and suppose that $\langle f_n \rangle$ converges uniformly on A to a function $f: A \to R$. Show that f is continuous on A. (6.5)

5(a) Let $f_n(x) = \frac{nx}{1+n^2x^2}$ for $x \geq 0$. Show that sequence $\langle f_n \rangle$ converges non-uniformly on $[0,\infty)$ and converges uniformly on $[a,\infty), a > 0$. (6.5)

(b) State and prove Weierstrass M-test for the uniform Convergence of a series of functions. (6.5)

(c) Show that the series of functions $\sum \frac{1}{n^2+x^2}$, converges uniformly *on* R to a continuous function. (6.5)

6(a) (i) Find the exact interval of convergence of the power series (3)

$$\sum_{n=0}^{\infty} 2^{-n} x^{3n}$$

(ii) Define $\sin x$ as a power series and find its radius of convergence (3)

(b) Prove that $\sum_{n=1}^{\infty} n^2 x^n = \frac{x(x+1)}{(1-x)^3}$ for $|x| < 1$ and hence evaluate $\frac{\sum_{n=1}^{\infty} n^2(-1)^n}{3^n}$. (6)

(c) Let $f(x) = \sum_{n=0}^{\infty} a_n x^n$ have radius of convergence $R > 0$. Then f is differentiable on $(-R,R)$ and

$$f'(x) = \sum_{n=1}^{\infty} n a_n x^{n-1} \quad for\ |x| < R. \qquad (6)$$

Name of the Course	: **B.Sc. (Hons.) Mathematics CBCS**
Semester	: **IV**
Unique Paper Code	: **32351402_OC**
Name of the Paper	: **C9 - Riemann Integration and Series of Functions**
Duration: **3 Hours**	Maximum Marks: **75**

Attempt any four questions. All questions carry equal marks. All symbols have usual meaning.

1. Calculate the upper Darboux sum and lower Darboux sum of the function $f(x) = \frac{1}{x^2}$ $\quad and\ g(x) = xe^x$ on the interval $[1, 2]$ for a partition $P = \{1, 1.25, 1.5, 1.75, 2\}$.

   Calculate upper and lower Darboux integral of $g(x) = x^2 + 5$ on $[2,4]$. Is g integrable ?

   Let $f: [a, b] \to R$ be a bounded function. Suppose that there is a partition P of $[a, b]$ such that
   $$L(f,\ P) = U(f,\ P).$$
   Show that f is a constant function.

2. Define $f(x) = x[x]$ on $[0, 4]$. Show that f is integrable, and evaluate $\int_0^4 f(x)dx$. Give an example where the functions f and g are not Riemann integrable, but $f.g$ is integrable. Let f be a continuous function on R and define
   $$F(x) = \int_0^{\frac{x^4}{4}} f(t)dt \quad for\ x \in R$$
   show that F is differentiable on R and compute F'.

3. Examine the convergence of following improper integrals
   $$\int_0^\infty e^{-x}(3x + 2)dx \qquad and \qquad \int_0^\infty \frac{dx}{\sqrt{3x^4 + 5x}}$$
   Using the properties of Gamma integral find the value of
   $$\int_0^\infty x^5 e^{-4x^2}\, dx$$

4. Let (f_n) be defined by $f_n(x) = 1 - |1 - x^2|^n\ \forall x \in [-\sqrt{2},\ \sqrt{2}]$
   Find the pointwise limit of (f_n) on $[-\sqrt{2},\ \sqrt{2}]$.
   Does the sequence converge uniformly on this interval? Justify your answer. Show that the sequence (f_n) where $f_n(x) = nxe^{-nx^2}$, $x \geq 0$ is not uniformly convergent on $[0,\ 2]$.
   Show that the sequence $\left\{\frac{\sin(n^2x^2+1)}{n(n+1)}\right\}$ converges uniformly on R.

5. Examine the convergence of the series of functions $\sum f_n$ where $f_n(x) = \frac{1}{1+x^n}$ and show that convergence is non uniform in $(1, \infty)$ and is uniform in $[a,\ \infty[,\ a > 1$

Show that $\sum_{n=1}^{\infty} \frac{\cos nx}{n^3}$ converges uniformly on R to a continuous function.

Evaluate the integral $\int_0^1 \sum_{n=1}^{\infty} \frac{x}{(n+x^2)^2} dx$

6. Find the radius of convergence and exact interval of convergence for the following power series

$$\sum_0^{\infty} \frac{4^n}{n5^{n+2}} (x-2)^n \qquad and \qquad \sum_0^{\infty} \left[\frac{3+5(-1)^n}{7}\right]^n x^n$$

Write the power series expansion for the integral of the following function:

$$f(x) = \frac{x^2}{3-x^3} \quad ,$$

given that $\frac{1}{1-x} = \sum_0^{\infty} x^n \quad , x \in]-1,1[.$

This question paper contains 4+2 printed pages]

Roll No. | | | | | | | | | | | |

S. No. of Question Paper : **2815**

Unique Paper Code : **32351402** **GC-4**

Name of the Paper : **C-9, Riemann Integration and Series of Functions**

Name of the Course : **B.Sc. (H) Mathematics**

Semester : **IV**

Duration : **3 Hours** Maximum Marks : **75**

(Write your Roll No. on the top immediately on receipt of this question paper.)

Attempt any *two* parts from each question.

All questions are compulsory.

1. (*a*) Prove that a bounded function f on $[a, b]$ is integrable if and only if for each $\varepsilon > 0$, $\exists$ a $\delta > 0$ such that :

$$\text{mesh}(P) < \delta \Rightarrow U(f, P) - L(f, P) < \varepsilon,$$

$\forall$ partitions P of $[a, b]$. 6

(*b*) Show that every continuous function on $[a, b]$ is integrable. Is the converse true ? Justify. 6

(*c*) State and prove Intermediate Value Theorem for integrals. Show, with the help of an example, that the condition of continuity cannot be relaxed. 6

P.T.O.

2. (*a*) (*i*) Let f be a bounded function on $[a, b]$. Define the Darboux Integral $\int_a^b f$. 3

(*ii*) Give an example of a function f which is not integrable, for which $|f|$ is integrable. 3

(*b*) (*i*) State Fundamental Theorem of Calculus-II. 2

(*ii*) Let f be a continuous function on $\mathbf{R}$, and define :

$$F(x) = \int_{x-1}^{x+1} f(t)\, dt \quad \text{for } x \in \mathbf{R}$$

Show that F is differentiable on $\mathbf{R}$, and compute F′. 4

(*c*) (*i*) Define :

$$f(x) = [x] \text{ on } [0, 3].$$

Show that f is integrable, and evaluate $\int_0^3 f(x)\, dx$. 3

(*ii*) Define :

$$g : [0, 2] \to [0, 4] \text{ as } g(x) = x^2.$$

Let :

$$P = \{0, \frac{1}{2}, 1, \frac{3}{2}, 2\}.$$ 3

Find U(g, P).

(a) Show that :

$$\int_0^{\infty} e^{-t} t^{x-1}\, dt$$

converges $\Leftrightarrow x > 0$. 7

(b) Determine the convergence or divergence of :

$$\int_0^{\infty} \frac{1}{e^x \sqrt{x}}\, dx.$$ 7

(c) In the following, determine the values of r for which the integral converges/diverges. Determine the values of integrals in case of convergence.

(i) $\int_1^{\infty} x^{-r}\, dx$

(ii) $\int_0^{\infty} x^{-r}\, dx$. 7

(a) Let $\{f_n\}$ be a sequence of integrable functions on $[a, b]$ and suppose that $\{f_n\}$ converges uniformly on $[a, b]$ to f. Show that f is integrable. 6

(b) Let

$$f_n = \frac{1}{(1+x)^n}$$

for x in $[0, 1]$. Find the point-wise limit f of the sequence $\{f_n\}$ on $[0, 1]$.

Does $\{f_n\}$ converge uniformly to f on $[0, 1]$? Justify. 6

P.T.O.

(c) Show that if $0 < b < 1$, then the convergence of the sequence :

$$f_n = \frac{x^n}{(1+x^n)}$$

for x in R, $x \geq 0$ is uniform on the interval [0, b], but is not uniform on the interval [0, 1]. 6

5. (a) State and prove Weierstrass M-Test and hence examine the uniform convergence of the series :

$$\sum_{n=1}^{\infty} \frac{1}{(x^2+n^2)}, x \text{ in } \mathbf{R}.$$ 6

(b) Show that the series of functions :

$$\sum \frac{1}{(1+x^n)}, x \neq 0,$$

converges uniformly on $[a, \infty)$ for $a > 1$, but is not uniformly convergent on $(1, \infty)$. 6

(c) Let $f : A \rightarrow \mathbf{R}$, $A \subseteq \mathbf{R}$ be a bounded function on A. Define uniform norm $\| f \|$ on A. State criterion for uniform convergence of a sequence of bounded functions using uniform norm and hence examine for uniform convergence the sequence :

$$f_n = x^n;\ x \in [0, 1].$$ 6

(a) (i) Given :

$$\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n$$

for $|x| < 1$. Evaluate $\sum_{n=1}^{\infty} \frac{n^2}{2^n}$. 2.5

(ii) Let

$$f(x) = \sum a_n x^n$$

for $|x| < R$. If $f(x) = f(-x), \forall |x| < R$, then show that $a_n = 0$ for all odd values of n. 4

(b) (i) If the power series $\sum_{n=0}^{\infty} a_n x^n$ has radius of convergence R, then show that series :

$$\sum_{n=1}^{\infty} n a_n x^{n-1} \text{ and } \sum_{n=0}^{\infty} \frac{a_n x^{n+1}}{n+1}$$

also have the radius of convergence R. 4

(ii) Show that :

$$f(x) = \sum_{n=0}^{\infty} \frac{x^n}{n!} \text{ for } x \in \mathbf{R}$$

is well-defined. Also show that : 2.5

$$f(x) = f'(x), \text{ for } x \in \mathbf{R}.$$

(c) (i) Define interval of convergence for a power series. Find the radius of convergence and interval of convergence for the following power series :

$$\sum_{n=1}^{\infty} \frac{x^n}{n}, \quad \sum_{n=1}^{\infty} \frac{x^n}{n^2}.$$

(ii) State Weierstrass's Approximation Theorem.

This question paper contains 4+2 printed pages]

Roll No. | | | | | | | | | | | |

S. No. of Question Paper : 2815

Unique Paper Code : 32351402 GC-4

Name of the Paper : C-9, Riemann Integration and Series of Functions

Name of the Course : B.Sc. (H) Mathematics

Semester : IV

Duration : 3 Hours Maximum Marks : 75

(Write your Roll No. on the top immediately on receipt of this question paper.)

Attempt any *two* parts from each question.

All questions are compulsory.

1. (*a*) Prove that a bounded function f on $[a, b]$ is integrable if and only if for each $\varepsilon > 0$, $\exists$ a $\delta > 0$ such that :

$$\text{mesh}(P) < \delta \Rightarrow U(f, P) - L(f, P) < \varepsilon,$$

$\forall$ partitions P of $[a, b]$. 6

(*b*) Show that every continuous function on $[a, b]$ is integrable. Is the converse true ? Justify. 6

(*c*) State and prove Intermediate Value Theorem for integrals. Show, with the help of an example, that the condition of continuity cannot be relaxed. 6

P.T.O.

2. (a) (i) Let f be a bounded function on $[a, b]$. Define the Darboux Integral $\int_a^b f$. 3

(ii) Give an example of a function f which is not integrable, for which $|f|$ is integrable. 3

(b) (i) State Fundamental Theorem of Calculus-II. 2

(ii) Let f be a continuous function on $\mathbf{R}$, and define :

$$F(x) = \int_{x-1}^{x+1} f(t)\, dt \quad \text{for } x \in \mathbf{R}$$

Show that F is differentiable on $\mathbf{R}$, and compute F′. 4

(c) (i) Define :

$$f(x) = [x] \text{ on } [0, 3].$$

Show that f is integrable, and evaluate $\int_0^3 f(x)\, dx$. 3

(ii) Define :

$$g : [0, 2] \to [0, 4] \text{ as } g(x) = x^2.$$

Let :

$$P = \{0, \frac{1}{2}, 1, \frac{3}{2}, 2\}.$$

Find $U(g, P)$. 3

3. (a) Show that :

$$\int_0^{\infty} e^{-t} t^{x-1} \, dt$$

converges $\Leftrightarrow x > 0$. 7

(b) Determine the convergence or divergence of :

$$\int_0^{\infty} \frac{1}{e^x \sqrt{x}} \, dx .$$ 7

(c) In the following, determine the values of r for which the integral converges/diverges. Determine the values of integrals in case of convergence.

(i) $\int_1^{\infty} x^{-r} \, dx$

(ii) $\int_0^{\infty} x^{-r} \, dx$. 7

4. (a) Let $\{f_n\}$ be a sequence of integrable functions on $[a, b]$ and suppose that $\{f_n\}$ converges uniformly on $[a, b]$ to f. Show that f is integrable. 6

(b) Let

$$f_n = \frac{1}{(1+x)^n}$$

for x in $[0, 1]$. Find the point-wise limit f of the sequence $\{f_n\}$ on $[0, 1]$.

Does $\{f_n\}$ converge uniformly to f on $[0, 1]$? Justify. 6

(c) Show that if $0 < b < 1$, then the convergence of the sequence :

$$f_n = \frac{x^n}{(1+x^n)},$$

for x in $\mathbf{R}$, $x \geq 0$ is uniform on the interval $[0, b]$, but is not uniform on the interval $[0, 1]$. 6

5. (a) State and prove Weierstrass M-Test and hence examine the uniform convergence of the series :

$$\sum_{n=1}^{\infty} \frac{1}{(x^2+n^2)}, x \text{ in } \mathbf{R}.$$

6

(b) Show that the series of functions :

$$\sum \frac{1}{(1+x^n)}, x \neq 0,$$

converges uniformly on $[a, \infty)$ for $a > 1$, but is not uniformly convergent on $(1, \infty)$. 6

(c) Let $f : A \to \mathbf{R}$, $A \subseteq \mathbf{R}$ be a bounded function on A. Define uniform norm $\| f \|$ on A. State criterion for uniform convergence of a sequence of bounded functions using uniform norm and hence examine for uniform convergence the sequence :

$$f_n = x^n;\ x \in [0, 1].$$

6

(a) (i) Given :

$$\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n$$

for $|x| < 1$. Evaluate $\sum_{n=1}^{\infty} \frac{n^2}{2^n}$. 2.5

(ii) Let

$$f(x) = \sum a_n x^n$$

for $|x| < R$. If $f(x) = f(-x), \forall |x| < R$, then show that $a_n = 0$ for all odd values of n. 4

(b) (i) If the power series $\sum_{n=0}^{\infty} a_n x^n$ has radius of convergence R, then show that series :

$$\sum_{n=1}^{\infty} n a_n x^{n-1} \text{ and } \sum_{n=0}^{\infty} \frac{a_n x^{n+1}}{n+1}$$

also have the radius of convergence R. 4

(ii) Show that :

$$f(x) = \sum_{n=0}^{\infty} \frac{x^n}{n!} \text{ for } x \in \mathbf{R}$$

is well-defined. Also show that :

$$f(x) = f'(x), \text{ for } x \in \mathbf{R}.$$ 2.5

(c) (i) Define interval of convergence for a power series. Find the radius of convergence and interval of convergence for the following power series :

$$\sum_{n=1}^{\infty} \frac{x^n}{n}, \quad \sum_{n=1}^{\infty} \frac{x^n}{n^2}.$$

(ii) State Weierstrass's Approximation Theorem. 15

[This question paper contains 4 printed pages.]

Your Roll No..............

Sr. No. of Question Paper : 1567 F-8

Unique Paper Code : 2351401

Name of the Paper : ANALYSIS III (Riemann Integration and Series of Functions)

Name of the Course : B.Sc. (H.) Mathematics

Semester : IV

Duration : 3 Hours Maximum Marks : 75

Instructions for Candidates

1. Write your Roll No. on the top immediately on receipt of this question paper.
2. Attempt any **two** parts from each question.
3. **All** questions are compulsory.

1. (a) Let f be a bounded function on [a, b]. Show that $L(f) \le U(f)$, where L(f) is the lower Darboux integral and U(f) is the upper Darboux integral on [a,b]. (6)

   (b) Let f be a continuous function on R and define $F(x)=\int_{x-1}^{x+1} f(t)dt$ for $x \in R$. Show that F is differentiable function on R. Compute F′. (6)

P.T.O.

(c) Suppose f and g are continuous function that $g(x) \geq 0$ for all $x \in [a,b]$. Prove th c in [a, b] such that :

$$\int_a^b f(t)g(t)\,dt = f(c)\int_a^b g(t)\,dt.$$

2. (a) Prove that every monotonic function integrable.

(b) Let f be defined on [0, b] as $f(x) = \begin{cases} x, \\ 0, \end{cases}$

Calculate upper and lower Darboux inte [0, b]. Is f integrable on [0, b] ?

(c) Let f be a bounded function on [a,b] If partitions of [a,b] such that $P \subseteq Q$, where exactly one point extra than the points of that $L(f, P) \leq L(f, Q) \leq U(f, Q) \leq U(f, P$

3. (a) Show that the improper integral $\int_1^\infty \frac{\sin t}{t}\,dt$ but not absolutely convergent.

(b) Show that the improper integral $\int_0^1 x^{m}$ converges if and only if $m > 0$, $n > 0$.

(c) Examine the convergence of the followi integrals :

(i) $\int_1^{\infty} \frac{1}{x\sqrt{x+1}}\,dx$

(ii) $\int_1^{\infty} e^{t^2}\,dt$ (3+3)

4. (a) Let $\langle f_n \rangle$ be a sequence of continuous functions on $A \subseteq R$ and suppose that (f_n) converges uniformly on A to $f : A \to R$. Then show that f is continuous on A. (6½)

(b) Let $f_n(x) = \frac{nx}{1+n^2x^2}$ for $x \geq 0$. Show that the sequence $\langle f_n \rangle$ converges only pointwise on $[0, \infty[$ and converges uniformly on $[a, \infty[$, $a > 0$. (6½)

(c) Let $f_n(x) = \frac{x}{1+nx^2}$ for $x \in R$, $n \in N$. Show that the sequence $\langle f_n \rangle$ converges uniformly on R. (6½)

5. (a) State and prove Weierstrass-M test for the uniform convergence of a series of functions. (2,5)

(b) Show that the series of function $\sum_{n=1}^{\infty} \sin\left(\frac{x}{n^2}\right)$ converges uniformly for $|x| \leq a$, $a > 0$ and is not uniformly convergent on R. (7)

(i) $\int_1^\infty \frac{1}{x\sqrt{x+1}}\,dx$

(ii) $\int_1^\infty e^{t^2}\,dt$ (3+3)

4. (a) Let $\langle f_n \rangle$ be a sequence of continuous functions on $A \subseteq R$ and suppose that (f_n) converges uniformly on A to $f : A \to R$. Then show that f is continuous on A. (6½)

(b) Let $f_n(x) = \frac{nx}{1+n^2x^2}$ for $x \geq 0$. Show that the sequence $\langle f_n \rangle$ converges only pointwise on $[0, \infty[$ and converges uniformly on $[a, \infty[$, $a > 0$. (6½)

(c) Let $f_n(x) = \frac{x}{1+nx^2}$ for $x \in R$, $n \in N$. Show that the sequence $\langle f_n \rangle$ converges uniformly on R. (6½)

5. (a) State and prove Weierstrass-M test for the uniform convergence of a series of functions. (2,5)

(b) Show that the series of function $\sum_{n=1}^{\infty} \sin\left(\frac{x}{n^2}\right)$ converges uniformly for $|x| \leq a$, $a > 0$ and is not uniformly convergent on R. (7)

P.T.O.

S Copy

(c) Show that the sequence $< x^2 e^{-nx} >$ c...
on $[0, \infty[$.

6. (a) Let $S(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots$ and $C(x)$...
$x \in R$. Prove that :

(i) $S' = C$ and $C' = -S$

(ii) $(S^2 + C^2)' = 0$

(iii) $S^2 + C^2 = 1$.

(b) Show that the function $E(x) = \sum_{n=...}^{\infty}$...
defined and is differentiable on R w...
$E'(x) = E(x)$ for all $x \in R$ and $E(x$...
all $x, y \in R$.

(c) Find the radius of convergence of th...
series :

(i) $\sum_{n=0}^{\infty} \frac{(n!)^2}{(2n)!} x^n$

(ii) $\sum_{n=1}^{\infty} n^{-\sqrt{n}} x^n$

...ed by

...basis

...matrix

...5,6.5)

This question paper contains 4 printed pages.

Your Roll No. ....

S. No. of Paper	: **6633**	**HC**
Unique paper code	: **32351402**	
Name of the paper	: **Reimann Integration and Series of Functions**	
Name of course	: **B.Sc. (Hons.) Mathematics**	
Semester	: **IV**	
Duration	: **3 hours**	
Maximum marks	: 75	

(Write your Roll No. on the top immediately on receipt of this question paper.)

*Attempt any **two** parts from each question.*
***All** questions are compulsory.*

Q.1 (a) Prove that a bounded function f on [a, b] is integrable if and only if for each $\varepsilon > 0, \exists$ a partition P of [a, b] such that $U(f, P) - L(f, P) < \varepsilon$. (6)

(b) Show that if f is integrable on [a, b], then $|f|$ is integrable on [a, b], and:

$$\left|\int_a^b f\right| \le \int_a^b |f|.$$

Hence, show that :

$$\left|\int_{-2\pi}^{2\pi} x^2 Sin^8(e^x)dx\right| \le \frac{16\pi^3}{3} \quad . \qquad (6)$$

(c) Suppose f and g are continuous functions on [a, b], and $g(x) \ge 0 \quad \forall x \in [a, b]$. Prove that, $\exists$ $x \in [a, b]$ such that:

$$\int_a^b f(t)g(t)\,dt = f(x)\int_a^b g(t)\,dt \,. \qquad (6)$$

Q.2 (a) (i) The Dirichlet function $f : [0, 1] \to \mathbf{R}$ is defined by :

$$f(x) = \begin{cases} 1, & x \in [0,1] \cap \mathbb{Q} \\ 0 & x \in [0,1] \setminus \mathbb{Q} \end{cases}$$

P. T. O.

Is f Riemann Integrable? Justify.

(ii) Show that a decreasing function on [a, b] is in[tegrable]. (

(b) (i) State Fundamental Theorem of Calculus-I. (

(ii) Let f be an integrable function on [a, b]. Let P be [a] partition of [a,b] and P* be a refinement of P such that P* [=] $P \cup \{u\}$. Show that $L(f, P) \leq L(f, P^*)$. (

(c) (i) For a bounded function f on [a, b], define th[e] Riemann Sum associated with a partition P. Hence, giv[e] Riemann's definition of integrability. (3

(ii) Calculate $\lim\limits_{h\to 0}\frac{1}{h}\int_{3}^{3+h} e^{t^2}\,dt$ (3

3. (a) Show that:

$$\beta(x,y)=\int_{0}^{1} t^{x}(1-t)^{y}\,dt \text{ converges } \Leftrightarrow x>0, y>0 .$$ (

(b) (i) Define Improper Integral of Type-II. When does [it] converge? When is it said to diverge ? (

(ii) Determine if $\int_{-\infty}^{\infty} e^{x}\,dx$ is a convergent integral. Justi[fy]. (

(c) (i) Determine if the following integrals are Improp[er] Integrals. If so, what kind ? Justify.

$$\int_{0}^{\pi/2}\frac{Sinx}{\sqrt[3]{x}}\,dx, \qquad \int_{0}^{1} x\ln(x)\,dx$$ (

(ii) Prove that $\int_{\pi}^{\infty}\frac{Sinx}{x}\,dx$ converges absolutely. (

4. (a) State and prove Cauchy's criterion for uniform co[n]vergence for sequences of functions. (6

(b) (i) Is the sequence $\{f_n\}$ where :

$$f_n = \frac{1}{n}\sin(nx + n).$$

uniformly convergent on $\mathbf{R}$? Justify. (2)

(ii) Suppose a sequence $\{f_n\}$ converges uniformly to f on a set A, and further suppose that each f_n is bounded on A. Show that the limit function f is bounded on A. (4)

(c) Show that if a>0, then the sequence $\{n^2x^2e^{-nx}\}$ converges uniformly on the interval $[a, \infty)$, but that it does not converge uniformly on the interval $[0, \infty)$. (6)

5. (a) If f_n is continuous on $D \subseteq \mathbf{R}$, for each $n \in \mathbb{N}$ and if Σf_n converges to f uniformly on D, then f is continuous on D. (6)

(b) Show that the series of functions $\Sigma \dfrac{x^n}{(1+x^n)}$, $x \geq 0$, converges uniformly on $[0, a]$ for $0 < a < 1$, but is not uniformly convergent on $[0, 1)$. (6)

(c) A sequence $\{f_n\}$ converges uniformly to f on A_0, if for each $\varepsilon > 0$ there is a natural number $K(\varepsilon)$, depending only on ε, such that if $n \geq K(\varepsilon)$, then:

$$|f_n(x) - f(x)| < \varepsilon \quad \text{for all } x \in A_0.$$

Hence state a necessary and sufficient condition for a sequence $\{f_n\}$ to fail to converge uniformly on A_0 to f.

Apply this condition on sequence $f_n(x) = \dfrac{x}{n}$, for $x \in \mathbf{R}$ to examine for uniform convergence. (6)

6. (a) (i) Suppose that $f(x)=\sum_{n=0}^{\infty} a_n x^n$ has radius of convergence $R>0$. Then show that $\int_0^x f(t)\,dt=\sum_{n=0}^{\infty}\frac{a_n}{n+1}x^{n+1}$ for $|x|<R$ (5)

(ii) State when is a power series differentiable term by term. (1.5)

(b) (i) Apply Abel's Theorem to show :

$$\log_e 2 = 1-\frac{1}{2}+\frac{1}{3}-\frac{1}{4}\dots. \qquad (3.5)$$

(ii) Given $\frac{1}{1-x}=\sum_{n=0}^{\infty}x^n$ for $|x|<1$. Evaluate:

$$\sum_{n=1}^{\infty}\frac{n}{5^n}, \quad \sum_{n=1}^{\infty}\frac{n^2(-1)^n}{3^n}. \qquad (3)$$

(c) ~~(i)~~ Apply Ratio Test for series to show that radius of convergence R of the power series $\sum a_n x^n$ is given by $\lim_{n\to\infty}\left|\frac{a_n}{a_{n+1}}\right|$, whenever the limit exists. (3.5)

(ii) Find radius of convergence for :

$$\sum_{n=2}^{\infty}(\ln(n))^{-1}x^n, \quad \sum_{n=0}^{\infty}\frac{(n!)^3 x^n}{(3n)!} \qquad (3)$$

This question paper contains 4 printed pages]

Roll No. | | | | | | | | | | | |

S. No. of Question Paper : **2251**

Unique Paper Code : **32351402** **IC**

Name of the Paper : **Riemann Integration and Series of Functions**

Name of the Course : **B.Sc. (H) Mathematics**

Semester : **IV**

Duration : **3 Hours** Maximum Marks : **75**

(Write your Roll No. on the top immediately on receipt of this question paper.)

Attempt any *two* parts from each question.

All questions are compulsory.

1. (*a*) Find the upper and the lower Darboux integrals for $f(x) = 2x^3$ on [0, 1]. Is f integrable on [0,1] ? Justify. 6

(*b*) Prove that a bounded function f on $[a, b]$ is Riemann integrable if and only if it is Darboux Integrable and the value of integrals agree. 6

(*c*) Prove that if f is integrable on $[a, b]$ then $|f|$ is integrable on $[a, b]$ and $\left|\int_a^b f\right| \leq \int_a^b |f|$. What about the converse ? Justify your answer. 6

2. (*a*) Prove that every bounded piecewise monotonic function f on $[a, b]$ is integrable. 6.5

P.T.O.

(*b*) State Fundamental Theorem of Calculus II. Let f be defined as follows :

$$f(t) = \begin{cases} 0 & \text{for } t < 0 \\ t & \text{for } 0 \leq t \leq 1 \\ 4 & \text{for } t > 1 \end{cases}$$

Determine the function $F(x) = \int_0^x f(t)\,dt$. Discuss the continuity and differentiability of F and also calculate F′. 6.5

(*c*) Let f be a bounded function on $[a, b]$. If P and Q are partitions of $[a, b]$ such that $P \subseteq Q$, show that :

$L(f, P) \leq L(f, Q) \leq U(f, Q) \leq U(f, P)$. 6.5

3. (*a*) Examine the convergence of the following improper integrals :

(*i*) $\int_1^\infty \frac{e^{-\sqrt{x}}}{\sqrt{x}}\,dx$

(*ii*) $\int_{-1}^1 \frac{dx}{x^3}$. 6

(*b*) Prove that :

$$\int_0^\infty \frac{\sin x}{x}\,dx$$

is convergent. 6

(*c*) Examine the convergence of the improper integral :

$$\int_0^1 x^{a-1}(1-x)^{b-1}\,dx.$$ 6

4. (a) Let (f_n) be a sequence of functions on $[a, b]$ converging uniformly to f on $[a, b]$. Prove that f is integrable on $[a, b]$ and

$$\int_a^b f = \lim \int_a^b f_n.$$ 6.5

(b) Let $f_n(x) = \dfrac{nx}{1 + n^2x^2}$ $\forall\, x \in \mathbf{R}$.

(i) Show that (f_n) converges pointwise to $f(x) = 0$ on $\mathbf{R}$.

(ii) Does (f_n) converge uniformly to f on $[0, 1]$? Justify.

(iii) Prove that (f_n) converges uniformly to f on $[1, \infty[$. 6.5

(c) Let $f_n : [0, 1] \to \mathbf{R}$ be defined for $n \geq 2$ by

$$f_n(x) = \begin{cases} n^2 x & \text{for } 0 \leq x \leq 1/n \\ -n^2(x - 2/n) & \text{for } 1/n \leq x \leq 2/n \\ 0 & \text{for } 2/n \leq x \leq 1. \end{cases}$$

Show that the sequence (f_n) is not uniformly convergent. 6.5

5. (a) Let (f_n) be a sequence of continuous functions on a set $A \subseteq \mathbf{R}$. If (f_n) converges uniformly to f on A, prove that f is continuous on A. 6

(b) Prove that the series :

$$\sum_{n=1}^{\infty} \sin\left(\frac{x}{n^2}\right)$$

is uniformly convergent on $[-a, a]$, $\forall\ a > 0$, but is not uniformly convergent on $\mathbf{R}$. 6

(c) Prove that the series :

$$\sum \frac{\cos\left(x^2+1\right)}{n^3}$$

represents a continuous function on $\mathbf{R}$. 6

6. (a) Find the radius of convergence of the power series :

(i) $\sum_{n=1}^{\infty} \frac{2^n}{n^2} x^n$

(ii) $\sum_{n=1}^{\infty} x^{n!}$. 6.5

(b) Let $\sum a_n x^n$ has radius of convergence $R > 0$ and let :

$$f(x) = \sum a_n x^n \text{ for } |x| < R.$$

Then prove that f is differentiable on $(-R, R)$ and

$$f'(x) = \sum_{n=1}^{\infty} n a_n x^{n-1} \text{ for } |x| < R.$$ 6.5

(c) Prove that :

$$\sum_{n=1}^{\infty} n^2 x^n = \frac{x(x+1)}{(1-x)^3}$$

for $|x| < 1$. Hence evaluate :

$$\sum_{n=1}^{\infty} \frac{n^2}{3^n}.$$ 6.5